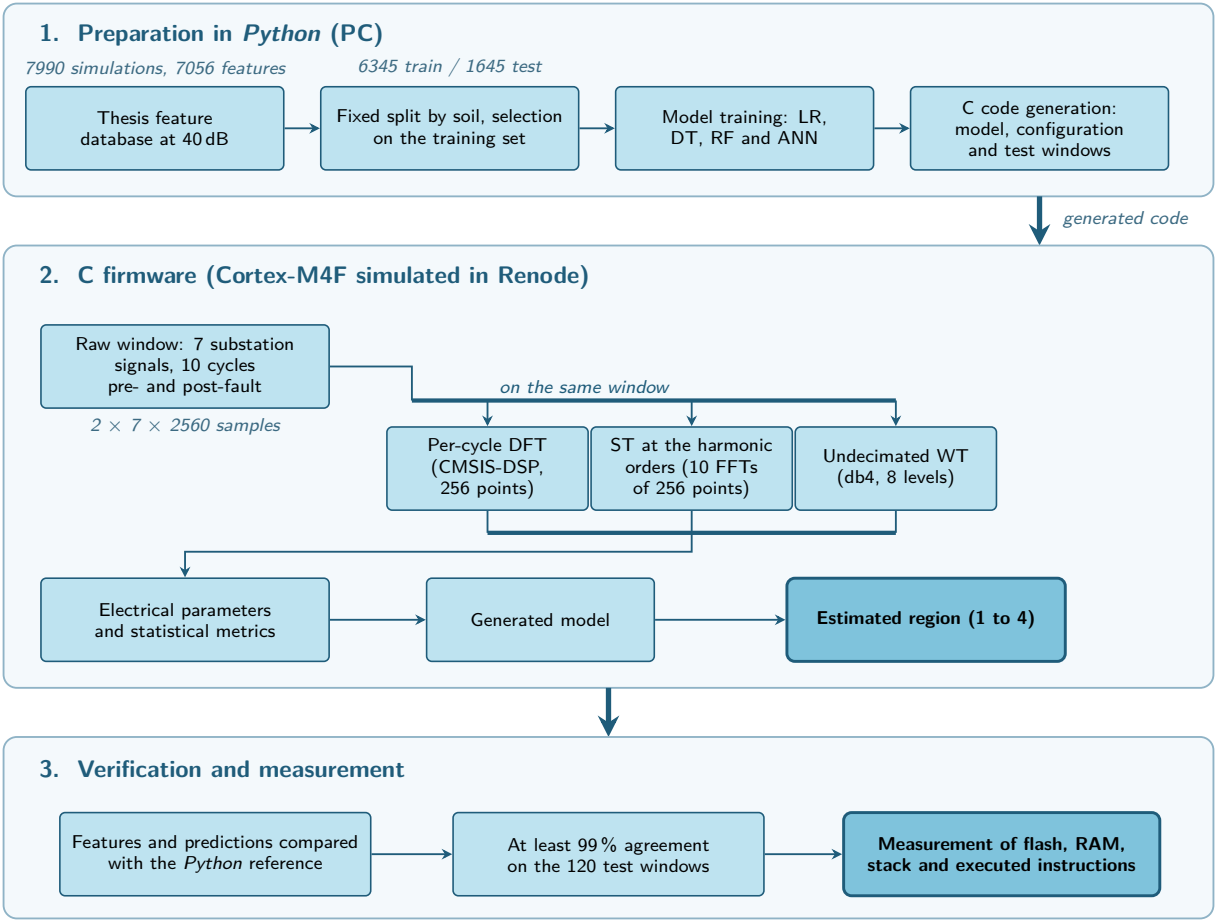

1. Preparation in Python (PC)
7990 simulations, 7056 features
6345 train / 1645 test
Thesis feature database at 40 dB
Fixed split by soil, selection on the training set
Model training: LR, DT, RF and ANN
C code generation: model, configuration and test windows
generated code
2. C firmware (Cortex-M4F simulated in Renode)
Raw window: 7 substation signals, 10 cycles pre- and post-fault
2 × 7 × 2560 samples
on the same window
Per-cycle DFT (CMSIS-DSP, 256 points)
ST at the harmonic orders (10 FFTs of 256 points)
Undecimated WT (db4, 8 levels)
Electrical parameters and statistical metrics
Generated model
Estimated region (1 to 4)
3. Verification and measurement
Features and predictions compared with the Python reference
At least 99 % agreement on the 120 test windows
Measurement of flash, RAM, stack and executed instructions